

UCO Physical Rehabilitation dataset

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The proposed dataset, coined “UCO Physical Rehabilitation”, has been recorded from June to July 2022. A total of 27 subjects (7 females and 20 males), in a range of age from 23 to 60 years, were recorded indoors in a controlled environment in the [AVA group](#) at the University of Cordoba.

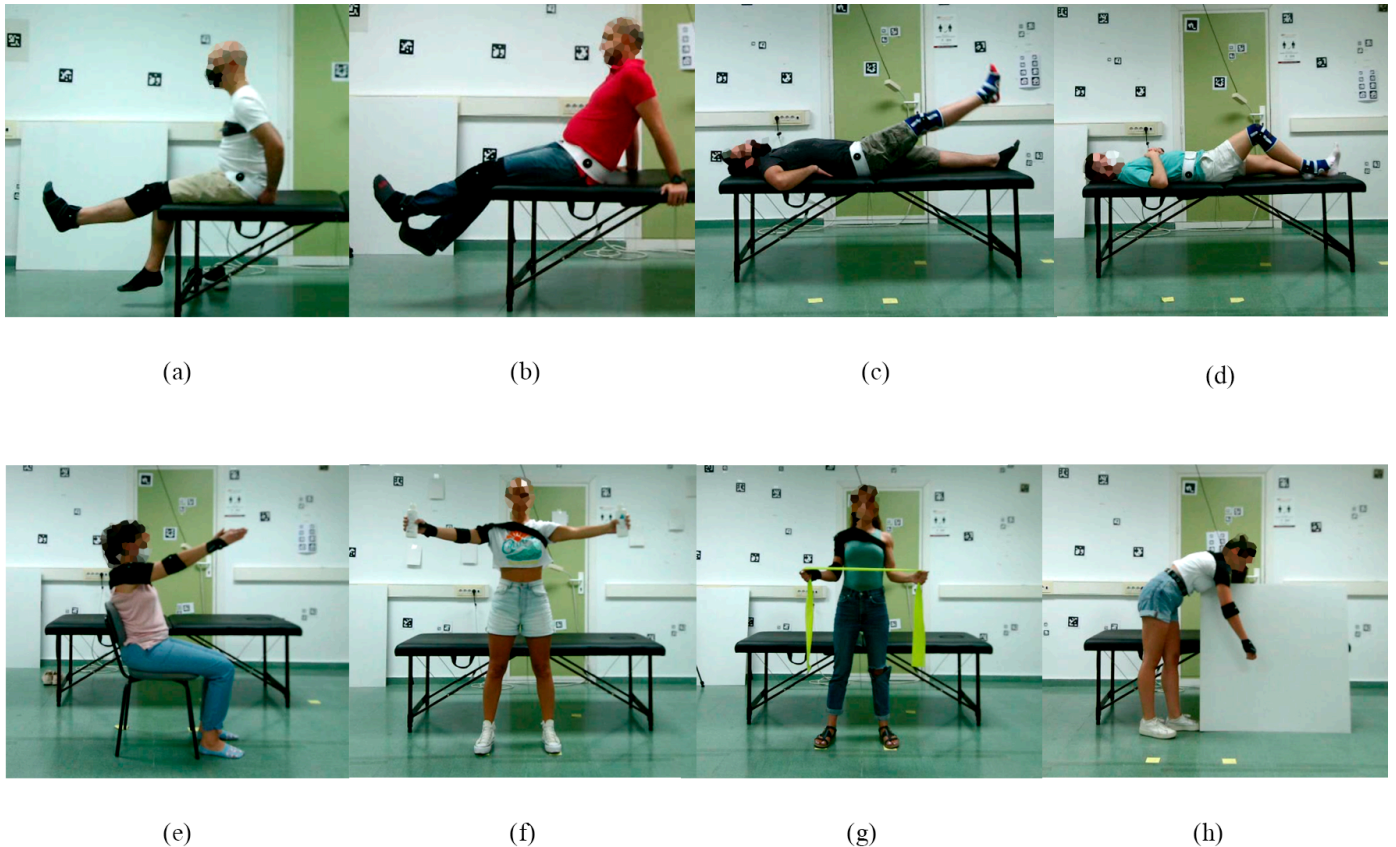


Figure 1 – Examples of some of the exercises included in the dataset: (a) Exercise 01, (b) Exercise 02, (c) Exercise 03, (d) Exercise 04, (e) Exercise 13, (f) Exercise 14, (g) Exercise 15, (h) Exercise 16. All views from the camera 2 viewpoint.

- Exercises 01 and 05 – Starting in a seated position, raise the leg as high as possible.
- Exercises 02 and 06 – In the same position as above, raise the leg as high as possible with the support of the other leg for raising and lowering.
- Exercises 03 and 07 – Lying flat on the treatment couch, raise the leg straight up as high as possible.
- Exercises 04 and 08 – In the same position as above, bend the knee with your heel on the treatment couch as much as possible.
- Exercises 09 and 13 – Seated in a chair, raise both arms straight as high as possible.
- Exercises 10 and 14 – Standing in front of the cameras, open and close both arms straight with a light weight.
- Exercises 11 and 15 – Standing in the same place as above, with elbows close to the body, stretch the rubber band with shoulder rotation.
- Exercises 12 and 16 – Standing in front of the treatment couch, with the other hand resting on the treatment couch, perform a pendulum shoulder rotation.

ID	Exercise name	Position	Side	Upper / Lower body
01	Bending the knee without support while sitting	Seated	Left	Lower
02	Bending the knee with support while sitting	Seated	Left	Lower
03	Lift the extended leg	Supine	Left	Lower
04	Bending the knee with bed support	Supine	Left	Lower
05	Bending the knee without support while sitting	Seated	Right	Lower
06	Bending the knee with support while sitting	Seated	Right	Lower
07	Lift the extended leg	Supine	Right	Lower
08	Bending the knee with bed support	Supine	Right	Lower
09	Shoulder flexion	Seated	Left	Upper
10	Horizontal weighted openings	Standing	Left	Upper
11	External rotation of shoulders with elastic band	Standing	Left	Upper
12	Circular pendulum	Standing	Left	Upper
13	Shoulder flexion	Seated	Right	Upper
14	Horizontal weighted openings	Standing	Right	Upper
15	External rotation of shoulders with elastic band	Standing	Right	Upper
16	Circular pendulum	Standing	Right	Upper

Table 1 – List of exercises included in the UCO Physical Rehabilitation dataset. They are classified by position, side and upper/lower body.

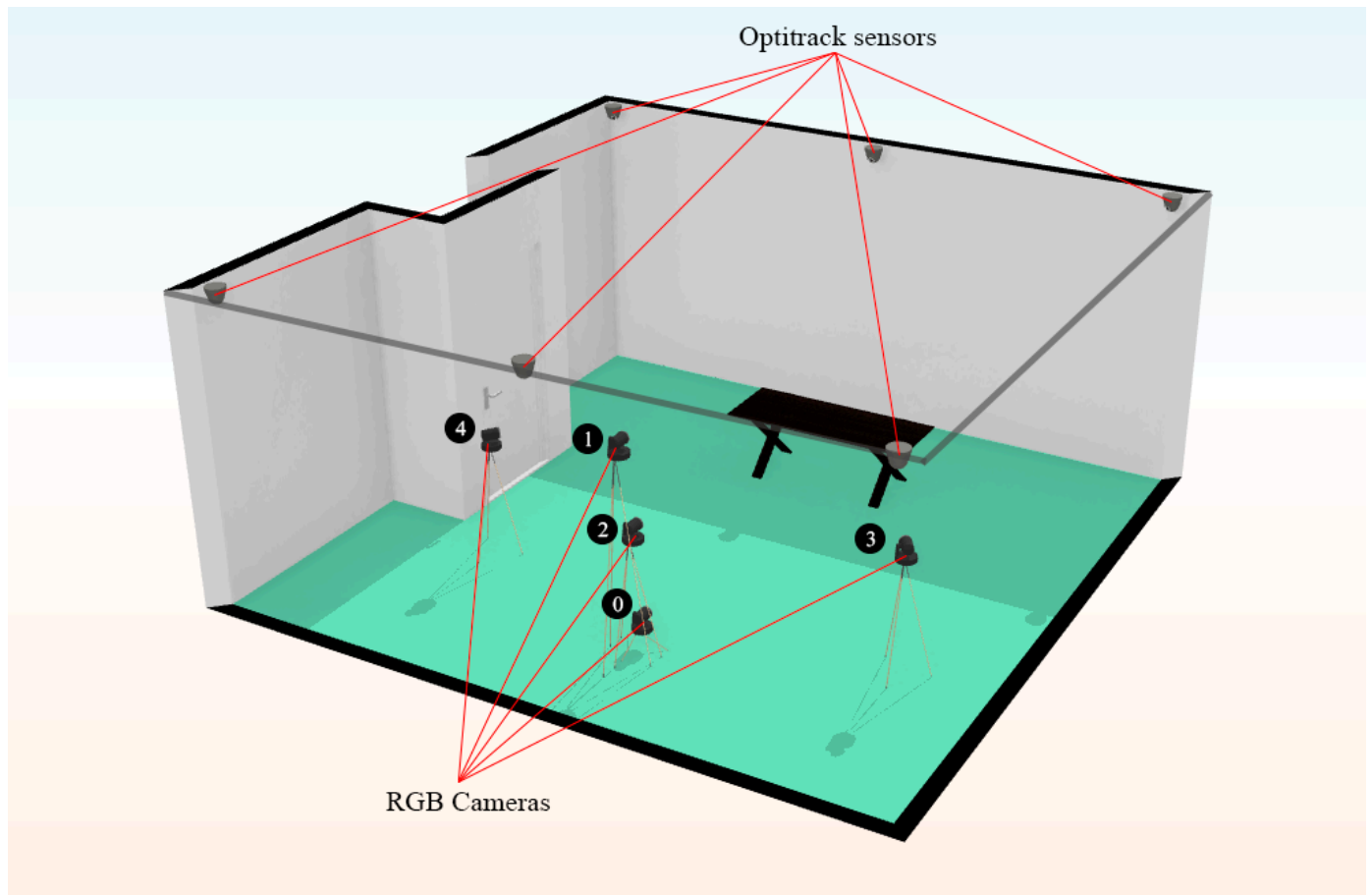


Figure 2 – Plan of the cameras in the laboratory. The scheme shows both the location of the OptiTrack motion capture system (solid blue lines) and the RGB cameras (solid red lines).

At the same time, the scene was recorded with an IR motion capture system (OptiTrack) with 6 cameras placed in an elevated position, as we can see in the Figure 2. Subjects were fitted with infrared sensors on the shoulder, elbow and wrist for upper body exercises and on the hip, knee and ankle for the lower body exercises. These IR sensors provide the 3D ground-truth points for each record on the dataset. Subsequently, we obtain the 2D ground-truth points from these 3D ground-truth points by projection.

Dataset structure

The raw data and video sequences are stored in the `data` folder. This folder contains the following subfolders and files:

```
.  
+-- dataset_2d.json  
+-- dataset_3d.json
```



```
| | | +-- cam1.avi
| | | +-- cam1_p2d.txt
| | | +-- cam2.avi
| | | +-- cam2_p2d.txt
| | | +-- cam3.avi
| | | +-- cam3_p2d.txt
| | | +-- cam4.avi
| | | +-- cam4_p2d.txt
| | | +-- p3d.txt
| | +-- 02
| | +-- ...
| | +-- 16
| +-- 1
| +-- ...
| +-- 27
```

The first level of subfolders corresponds with all of the 27 individuals participants in the dataset. Each individual folder contains 16 subfolders, one for each exercise they performed. In each exercise folder, we can find a `camX.avi` and a `camX_p2d.txt` file for each camera and a `p3d.txt` file.

The `camX_p2d.txt` contains the joints coordinates

dataset_2d.json

In Figure 3 you can see the structure of the dataset JSON for 2D. This is the first individual (folder), the first camera and the first frame of the record.



```
8      "body": "lower",
9      "cameras": {
10        "0": {
11          "frames": [
12            {
13              "id": 0,
14              "joints": {
15                "l_hip": {
16                  "x": "575.896",
17                  "y": "429.421"
18                },
19                "l_knee": {
20                  "x": "421.929",
21                  "y": "454.204"
22                },
23                "l_ankle": {
24                  "x": "368.39",
25                  "y": "573.135"
26                }
27              }
28            },
```

Figure 3 – Structure of the dataset JSON for 2D.

dataset_3d.json

In Figure 4 you can see the structure of the dataset JSON for 3D. This is the first individual (folder) and the first frame of the record.



```
8      "body": "lower",
9      "frames": [
10         {
11             "id": 0,
12             "joints": {
13                 "l_hip": {
14                     "x": "1.2086",
15                     "y": "-0.65282",
16                     "z": "-1.28515"
17                 },
18                 "l_knee": {
19                     "x": "0.785226",
20                     "y": "-0.554628",
21                     "z": "-1.61831"
22                 },
23                 "l_ankle": {
24                     "x": "0.703331",
25                     "y": "-0.148558",
26                     "z": "-1.78596"
27                 }
28             }
29         }
30     ]
31 }
```

Figure 4 – Structure of the dataset JSON for 3D.

Citation

```
@Article{aguilar_2023,
AUTHOR = {Aguilar-Ortega, Rafael and Berral-Soler, Rafael and Jiménez-Vela
TITLE = {UCO Physical Rehabilitation: New Dataset and Study of Human Pose
JOURNAL = {Sensors},
VOLUME = {23},
YEAR = {2023},
NUMBER = {21},
ARTICLE-NUMBER = {8862},
URL = {https://www.mdpi.com/1424-8220/23/21/8862},
ISSN = {1424-8220},
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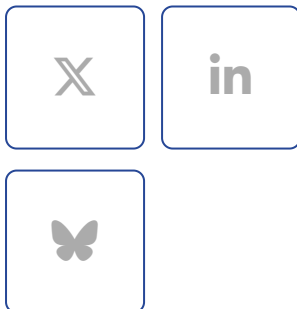


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